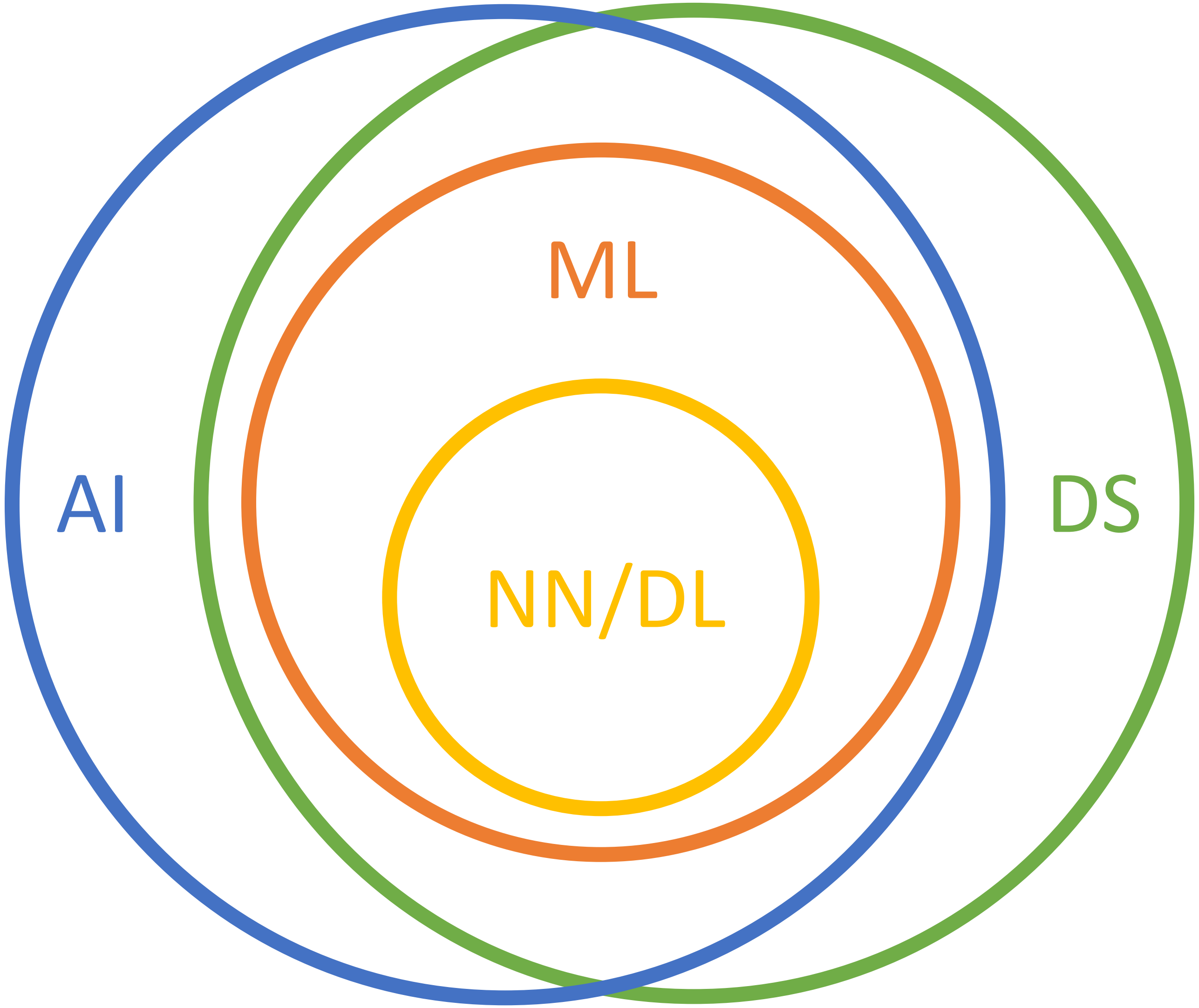
TYPES OF MACHINE LEARNING

MACHINE LEARNING VS. NEURAL NETWORKS



MACHINE LEARNING

- Improving performance of a system on future tasks based on observations/data about the population

- 3 types of machine learning:
  - Supervised learning
  - Unsupervised learning
  - Reinforcement learning

SUPERVISED LEARNING

- Data: Pairs (x, y)
  - x is input
  - y is label

- Objective: Learn a function f, such that f(x) = y

SUPERVISED LEARNING

- Analogy-based methods
  - Nearest neighbour
  - Support vector machine
- Inductive methods
  - Rule-based systems
  - Decision trees
- Connective methods
  - Neural networks (!!)
- Evolutionary methods
  - Genetic algorithm
- Bayesian methods
  - Bayesian inference

UNSUPERVISED LEARNING

- Data: x
  - x is input

- Objective: Learn structure of data

UNSUPERVISED LEARNING

- Clustering methods
  - Hierarchical clustering
  - K-means clustering
- Dimensionality reduction methods
  - Principal components analysis
  - Autoencoders
- Density estimation
  - Kernel density estimation

REINFORCEMENT LEARNING

- Data: s, r
  - s is environment state
  - r is reward

- Objective: Learn actions maximizing reward

PERFORMANCE MEASURES (SUPERVISED LEARNING)

CONFUSION MATRIX

		Prediction	
		0	1
Ground-Truth	0	true negatives	false positives
	1	false negatives	true positives

		Prediction	
		0	1
Ground-Truth	0	72	9
	1	6	13

BINARY CLASSIFICATION

$$Accuracy = \frac{\#\, correct\ predictions}{\#\, predictions}$$

$$Sensitivity = Recall = \frac{\#\, true\ positives}{\#\, positives}$$

$$Specificity = \frac{\#\, true\ negatives}{\#\, negatives}$$

$$Precision = \frac{\#\, true\ positives}{\#\, predicted\ positives}$$

$$F_1 = 2\left(\frac{precision \times recall}{precision + recall}\right)$$

MULTI-CLASS CLASSIFICATION

		Prediction			
		0	1	2	3
Ground-Truth	0	72	9	2	1
	1	6	13	3	0
	2	2	3	5	5
	3	0	4	11	4

“MACRO” AVERAGE

$$MacroAccuracy = \frac{1}{|classes|} \sum_{c \in classes} \frac{\#\ correct\ predictions\ for\ c}{\#\ predictions\ for\ c}$$

$$MacroSensitivity = MacroRecall = \frac{1}{|classes|} \sum_{c \in classes} \frac{\#\ true\ positives\ for\ c}{\#\ positives\ for\ c}$$

$$MacroSpecificity = \frac{1}{|classes|} \sum_{c \in classes} \frac{\#\ true\ negatives\ for\ c}{\#\ negatives\ for\ c}$$

$$MacroPrecision = \frac{1}{|classes|} \sum_{c \in classes} \frac{\#\ true\ positives\ for\ c}{\#\ predicted\ positives\ for\ c}$$

$$MacroF_1 = 2\left(\frac{macroprecision \times macrorecall}{macroprecision + macrorecall}\right)$$

"MICRO" AVERAGE

$$MicroAccuracy = \frac{\sum_{c \in classes} \#\, correct\ predictions\ for\ c}{\sum_{c \in classes} \#\, predictions\ for\ c}$$

$$MicroSensitivity = MicroRecall = \frac{\sum_{c \in classes} \#\, true\ positives\ for\ c}{\sum_{c \in classes} \#\, positives\ for\ c}$$

$$MicroSpecificity = \frac{\sum_{c \in classes} \#\, true\ negatives\ for\ c}{\sum_{c \in classes} \#\, negatives\ for\ c}$$

$$MicroPrecision = \frac{\sum_{c \in classes} \#\, true\ positives\ for\ c}{\sum_{c \in classes} \#\, predicted\ positives\ for\ c}$$

$$MicroF_1 = 2\left(\frac{microprecision \times microrecall}{microprecision + microrecall}\right)$$

REGRESSION

Ground-Truth	2.1	3.4	5.6	5.9	7.6	6.3	2.3	4.7	4.8	1.2
Prediction	1.5	4.4	5.1	4.8	6.3	6.2	3.1	4.8	5.6	3.2

REGRESSION

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

$$corr = \frac{cov(\hat{y}, y)}{\sigma_{\hat{y}}\sigma_y}$$

Carleton UNIVERSITY
Canada's Capital University

VALIDATION STRATEGIES (SUPERVISED LEARNING)

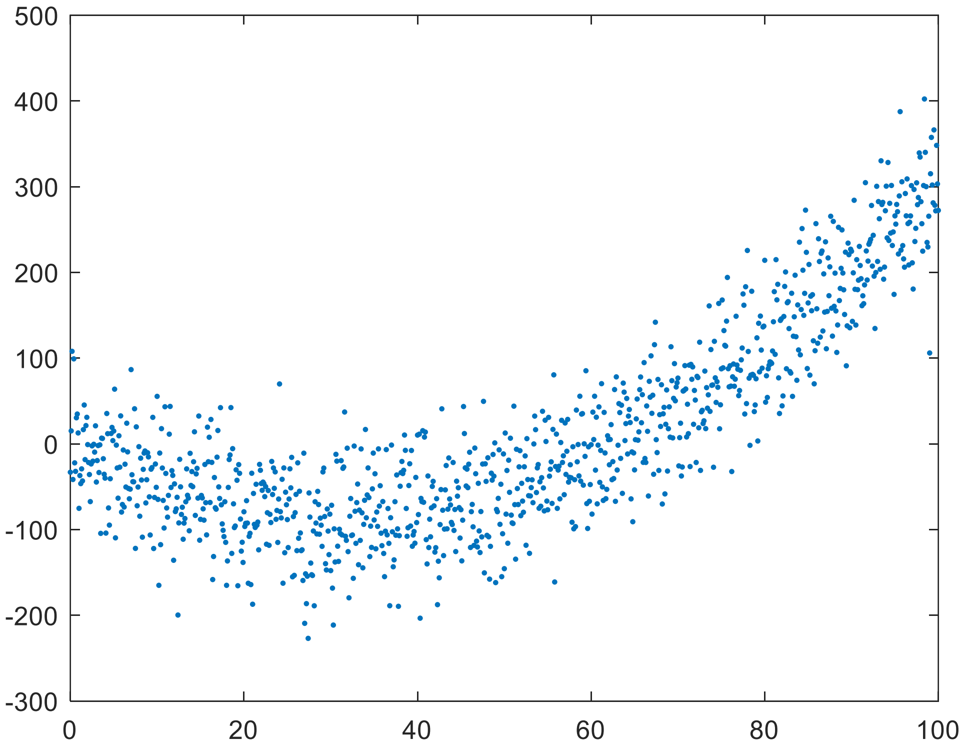
TRAIN/VALIDATION/TEST SETS

Training set: Fine tune model's parameters

Validation set: Fine tune model's hyperparameters

Testing set: Evaluate model's performance

MODEL VALIDATION EXAMPLE



$$y = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

HOLD-OUT VALIDATION

Assume we have N samples, called x_i.

Split x_i into 3 different sets: training, validation, test.

for θ in all combinations of hyperparameters:

  train model $M(\theta)$ on training set

  validate $M(\theta)$ on validation set

test $M(\theta)$ on test set using θ achieving best performance on validation set

LEAVE-ONE-OUT CROSS-VALIDATION

Assume we have N samples, called x_i.

for $i = 1 \ldots N$:

train model M on all samples x_k for $k \neq i$

test M on x_i

Performance is mean over all x_i

K-FOLD CROSS-VALIDATION

Assume we have N samples, called x_i.

Split x_i into k different folds f_i.

for $i = 1 \dots k$:

split remaining data into training set and validation set

for θ in all combinations of hyperparameters:

train model $M(\theta)$ on current training set

validate $M(\theta)$ on current validation set

test $M(\theta)$ on f_i using θ achieving best performance on validation set

test performance is mean over all test folds

K-FOLD CROSS-VALIDATION